Hoffmann's single-phase motor in Kron's frames

Kron's non-Riemannian electrodynamics (Hoffmann 1949): does the torsion of a non-holonomic frame carry the motor's torque? — CLAUDE.md §3a, M9c.1

kronspice simulation report · 2026-11-07 · 120 V, 60 Hz, 5 s runs

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Summary

B. Hoffmann, “Kron's Non-Riemannian Electrodynamics,” *Rev. Mod. Phys.* 21, 535–540 (1949), explains for physicists what Kron did with rotating machines and works one example, a single-phase squirrel-cage induction motor. This example builds that motor and solves it three ways: with rotor coils that turn with the rotor (frame I), with Kron's stationary rotor coils and speed matrix (frame II(b)), and by integrating Hoffmann's eq. 5′ with the torsion term directly. New in kronspice for this: *kron.Torsion* (eq. 19), *kron.NewMetric* (eq. 15) and *kron.NonRiemannian* (eq. 5′), in *kron/hoffmann.go*.

- **Constant inductances, but torsion:** in the stationary-coil frame every inductance is constant, and the frame has torsion Ω(2s θ 3s) = −0.125, Ω(3s θ 1) = +0.115, Ω(3s θ 2s) = +0.125, independent of θ (spread 10^{-16}).
- **Eq. 5′ is exact:** at 200 random states it gives the frame-I forces to 5×10^{-16}; without the torsion term it misses by up to 293 % of the largest force — Hoffmann's warning.
- **Three solutions agree:** pushed to +40 rad/s, the motor reaches 376.587 rad/s in frame I and by eq. 5′ (difference 4.1×10^{-4} rad/s), 376.618 rad/s in frame II(b).
- **No start from rest; runs either way when pushed:** from rest the final speed is 0; pushed to −40 rad/s it runs to −376.587 rad/s.
- **Without the torsion it only coasts:** $40\cdot e^{-5\ \mathrm{s}\cdot\mathrm{friction/J}}$ = 24.261 rad/s.
- **Every ledger closes** to 2×10^{-6} or better; every joule comes from the 120 V supply (or the push).

The idea

A machine's currents and its rotor angle together are the coordinates of one moving point. The inductances and the inertia act as the geometry (the metric) of the space it moves in, and Lagrange's equations are the path of the point (Hoffmann eqs. 1–8).

- **Frame I: coils that turn with the rotor** (Fig. 2). The squirrel cage is replaced by two perpendicular coils fixed to the rotor. Their inductances to the stator change with the angle θ (eqs. 24, 42). The space is ordinary (Riemannian) and the equations are Lagrange–Maxwell with Christoffel symbols (eq. 5) (*kron.Christoffel*, *kron.Geodesic*).
- **Stationary coils** (Hoffmann's barred 2 and 3, written 2s and 3s here; Fig. 3, eq. 35). Current paths that stay still while the cage turns make every inductance a constant (eq. 29). The price is that this change of frame is *non-holonomic*: no set of coordinates goes with it (eq. 38). The space gains a torsion Ω (eq. 19), and the equations gain a term $2\Omega\cdot\dot{x}\cdot\dot{x}$ (eq. 5′) that carries both the speed voltages and the torque.
- **Hoffmann's warning** (p. 540): putting the constant inductances into the ordinary Lagrange–Maxwell equations “would obtain wrong results”.

The motor (all values declared)

part	value	part	value
stator coil L11	0.25 H, 2 Ω	inertia J	0.001 kg·m²
rotor coils L_S	0.25 H, 3 Ω each	friction	10^{-4} N·m·s
mutual M	0.23 H	supply	120 V, 60 Hz
salient pole L_D	0 (and 0.01, 0.02 H in some runs)	poles; synchronous speed	two; 377.0 rad/s

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1. The two frames

In the stationary-coil frame the inductance matrix (henry; inertia in kg·m²) no longer depends on θ:

	1	2s	3s	θ
1	0.2500	0.2300	0.0000	0.0000
2s	0.2300	0.2500	0.0000	0.0000
3s	0.0000	0.0000	0.2500	0.0000
θ	0.0000	0.0000	0.0000	0.0010

The frame's torsion $\Omega_{\lambda\mu\nu}$ has non-zero components Ω(2s θ 3s) = −0.125, Ω(3s θ 1) = +0.115 and Ω(3s θ 2s) = +0.125, with Ω(μλν) = −Ω(λμν); it does not change with θ (spread 10^{-16}). At 200 random states (angles, currents, their rates, speed, acceleration), eq. 5′ with the torsion gives the frame-I forces to 5×10^{-16}. Ordinary Lagrange–Maxwell equations on the constant inductances miss by up to 293 % of the largest force.

2. Starting (5 s runs)

run	final speed	energy from the mains	ledger mismatch
from rest, frame I	**0.000 rad/s** — does not start	1409 J = heat 1407 J + stored 2.2 J	3×10^{-7}
pushed to +40 rad/s, frame I	**376.587 rad/s**	867 J = heat 796 J + stored 70.5 J	2×10^{-6}
pushed to +40 rad/s, frame II(b)	376.618 rad/s	867 J = heat 796 J + stored 70.6 J	1×10^{-7}
pushed to +40 rad/s, eq. 5′ (Runge–Kutta)	**376.587 rad/s**	—	—
pushed to −40 rad/s, frame I	−376.587 rad/s	867 J = heat 796 J + stored 70.5 J	2×10^{-6}
pushed to +40 rad/s, no torsion	24.261 rad/s — coasts on friction	1409 J = heat 1407 J + stored 1.7 J	3×10^{-7}

- **No start from rest.** A single-phase motor has no starting torque: its pulsating field is two equal fields turning in opposite directions. Real motors add a start winding or a capacitor.
- **Pushed, it runs either way**, up to just under synchronous speed (slip 0.107 % with no load).
- **The three solutions agree.** Frame I and eq. 5′ agree to 4.1×10^{-4} rad/s. Eq. 5′ is integrated by this example's own Runge–Kutta code, which shares nothing with kronspice's machine element.
- **Hoffmann's warning.** Without the torsion there is no speed voltage and no torque; the pushed rotor follows friction alone.

3. Torque–speed curve

With the shaft held at each speed, mean torque over the last 10 cycles (N·m). The curve is odd in speed (the negative half is the same with the sign changed).

rad/s	0.0	62.5	125.0	187.6	250.1	312.6	375.1
$L_D = 0$	0.0000	0.0707	0.1523	0.2619	0.4327	0.7121	0.0714
L_D = 0.02 H (salient pole)	0.0000	0.2309	0.4845	0.7854	1.1438	1.3829	0.0845

- **Shape.** Zero at standstill, odd in speed, largest near 91 % of synchronous speed. Frames I and II(b) agree to 1.7×10^{-3} of the peak.
- **Salient pole.** It makes stationary coil 2s couple more tightly to the stator (stationary-frame $L_{2s2s} = L_S - L_D = M$), which roughly doubles the torque at most speeds.
- **No-load slip** falls with L_D: 376.587 rad/s (0.107 %) at 0 H, 376.694 rad/s (0.079 %) at 0.01 H, 376.793 rad/s (0.053 %) at 0.02 H. The motor does not pull into synchronism at these values: a reluctance motor needs a much larger L_D.

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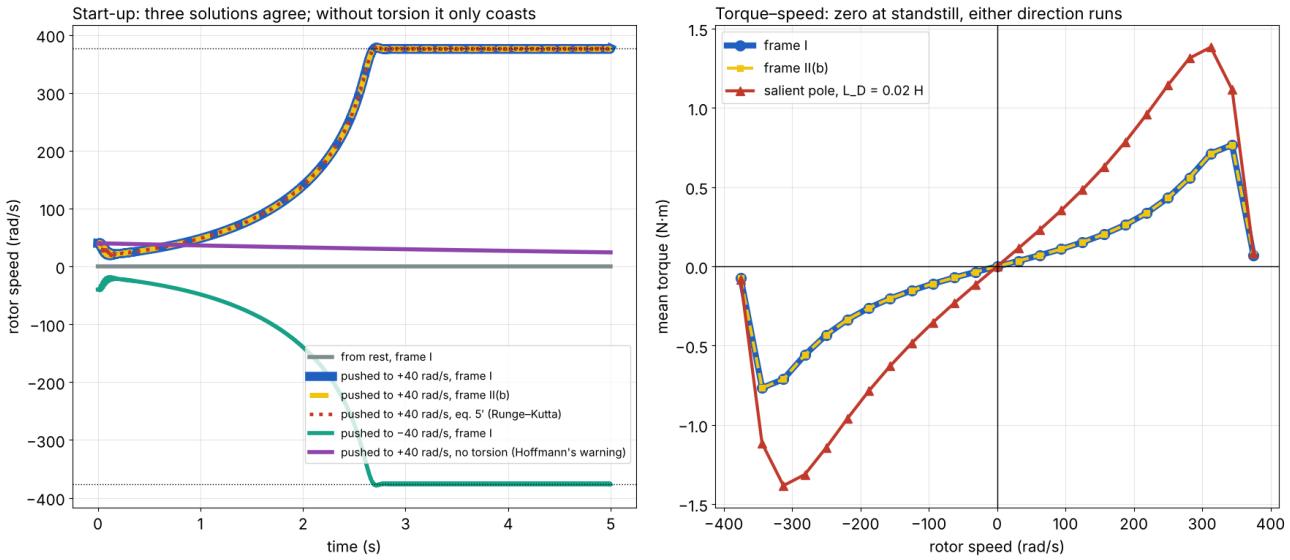


Figure 1. Left: rotor speed in the start-up runs. From rest the motor stays at 0; pushed to +40 rad/s, frame I, frame II(b) and eq. 5′ overlap and run up to just under synchronous speed (dotted lines); pushed to −40 rad/s it runs backwards; without the torsion term it only coasts down. Right: mean torque against held shaft speed, frames I and II(b) overlapping, and with a salient pole L_D = 0.02 H.

Validation

main_test.go checks the random-state twin, the constancy of the torsion, start-up agreement, the coast-down and the torque curve; *kron/hoffmann_test.go* tests *Torsion*, *NewMetric* and *NonRiemannian*. Every kronspice run's energy ledger closes to 2×10^{-6} or better. The motor adds no energy: every joule of shaft motion and heat comes from the 120 V supply (or the push).

Limits. All machine values are declared, not taken from a real motor; the rotor cage is represented by two coils, as in Hoffmann's paper; the motor runs with no load.

Reproducing the results

```
go run ./examples/hoffmann_motor -viz examples/hoffmann_motor/viz   # about 10 seconds
python3 examples/hoffmann_motor/viz/render.py
go test ./examples/hoffmann_motor ./kron
python examples/hoffmann_motor/viz/make_report.py examples/hoffmann_motor/hoffmann_motor_report.pdf
```

Sources: B. Hoffmann, Rev. Mod. Phys. 21, 535–540 (1949); G. Kron, “Non-Riemannian Dynamics of Rotating Electrical Machinery,” J. Math. Phys. 13, 103–194 (1934); G. Kron, Equivalent Circuits of Electric Machinery (Wiley, 1951), Epilogue — frames I and II(b) as used by kron.RotatingFrame.

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